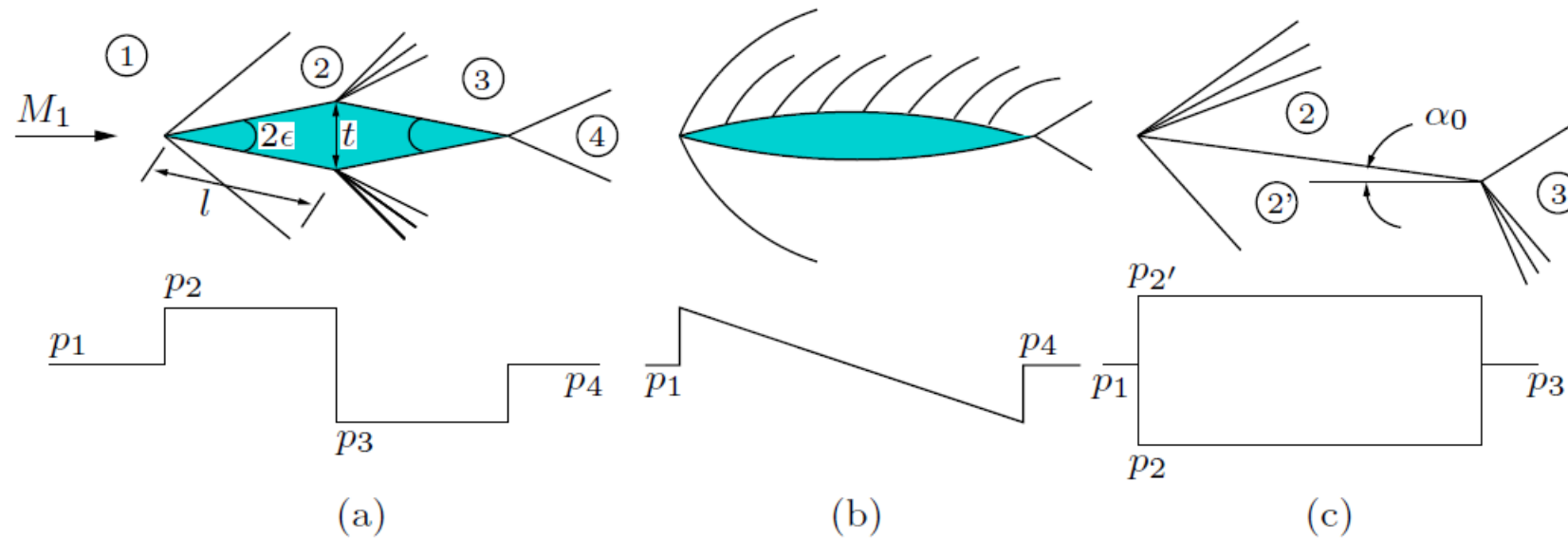


Shock-Expansion Theory

Aerodynamic forces acting on a body present in a supersonic flow are governed by the shock and expansion waves formed at the surface of the body

The pressure distribution over an object depends on the wave pattern on it



Diamond Airfoil /Double Wedge At zero angle of attack

Lift is zero

The drag per unit span

$$D = 2(p_2 l \sin \varepsilon - p_3 l \sin \varepsilon) = 2(p_2 - p_3)(t/2)$$

$$D = (p_2 - p_3) t$$

For the flat plate at an angle of attack α_0

$$L = (p_{2'} - p_2) c \cos \alpha_0$$

$$D = (p_{2'} - p_2) c \sin \alpha_0$$

Thin Aerofoil Theory

When the airfoil is thin and is kept at a small angle of attack, i.e. if the flow inclinations are small

weak oblique shocks

Pressure change across a weak shock

$$\frac{\Delta p}{p_1} \approx \frac{\gamma M_1^2}{\sqrt{M_1^2 - 1}} \Delta \theta$$

$$\frac{p - p_1}{p_1} = \frac{\gamma M_1^2}{\sqrt{M_1^2 - 1}} (\theta - 0)$$

$$C_p = \frac{p - p_1}{q_1} = \frac{2}{\gamma M_1^2} \frac{p - p_1}{p_1}$$

$$C_p = \frac{2\theta}{\sqrt{M_1^2 - 1}}$$

pressure coefficient is proportional to the local flow direction

For the flat plate at a small angle of attack –Thin airfoil theory

$$C_p = \mp \frac{2\alpha_0}{\sqrt{M_1^2 - 1}}$$

minus sign is for C_p on the upper surface and the plus sign is for C_p on the lower surface

Small alpha

$$C_L = (C_{pl} - C_{pu})$$

$$C_D = (C_{pl} - C_{pu})\alpha_0$$

$$C_L = \frac{(p_l - p_u) c \cos \alpha_0}{q_1 c} = (C_{pl} - C_{pu}) \cos \alpha_0$$

$$C_D = \frac{(p_l - p_u) c \sin \alpha_0}{q_1 c} = (C_{pl} - C_{pu}) \sin \alpha_0$$

$$C_L = \frac{4\alpha_0}{\sqrt{M_1^2 - 1}}$$

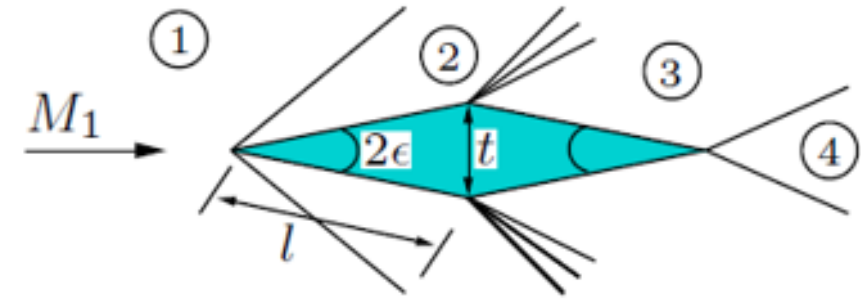
$$C_D = \frac{4\alpha_0^2}{\sqrt{M_1^2 - 1}}$$

Diamond section aerofoil

nose angle 2ϵ , at zero angle of attack.

$$C_p = \pm \frac{2\epsilon}{\sqrt{M_1^2 - 1}}$$

+ sign is for the front face where the pressure p_2 is higher than p_1 and the - sign is for the rear face with pressure p_3 less than p_1



$$p_2 - p_3 = \frac{4\epsilon}{\sqrt{M_1^2 - 1}} q_1$$

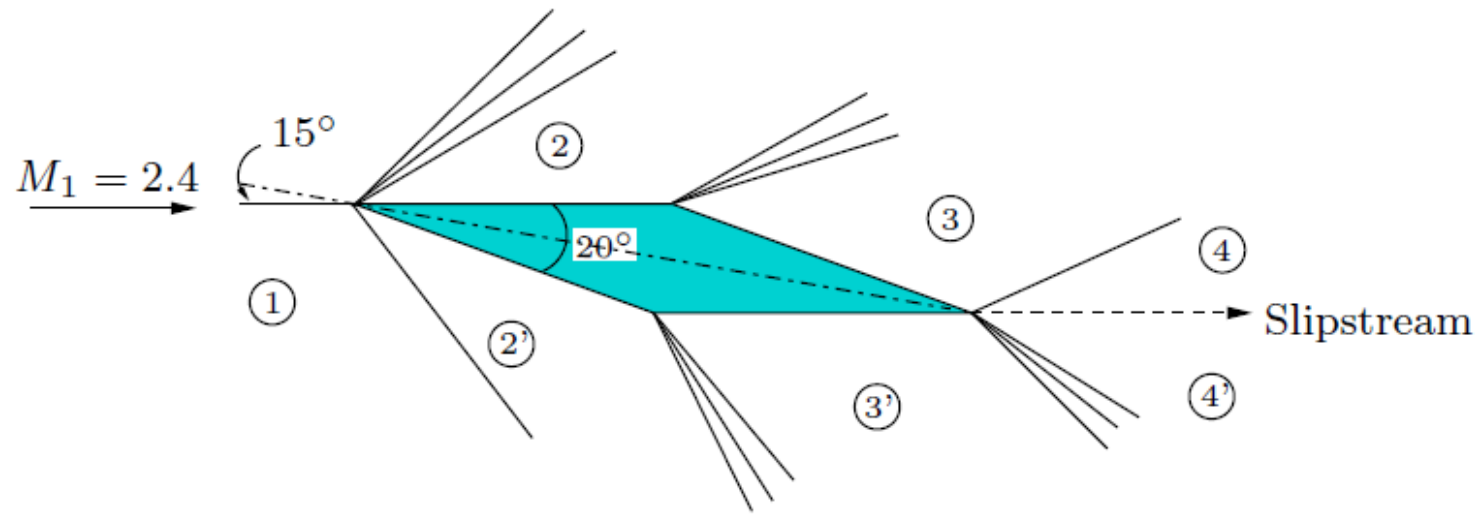
$$D = (p_2 - p_3)t = (p_2 - p_3)\epsilon c$$

$$D = \frac{4\epsilon^2}{\sqrt{M_1^2 - 1}} q_1 c$$

$$C_D = \frac{D}{q_1 c} = \frac{4\epsilon^2}{\sqrt{M_1^2 - 1}}$$

$$C_D = \frac{4}{\sqrt{M_1^2 - 1}} \left(\frac{t}{c} \right)^2$$

Determine the flow field around a symmetric double wedge of 20° included angle kept at 15° angle of attack to a Mach 2.4 air stream at a stagnation temperature of 300 K



Without slip stream

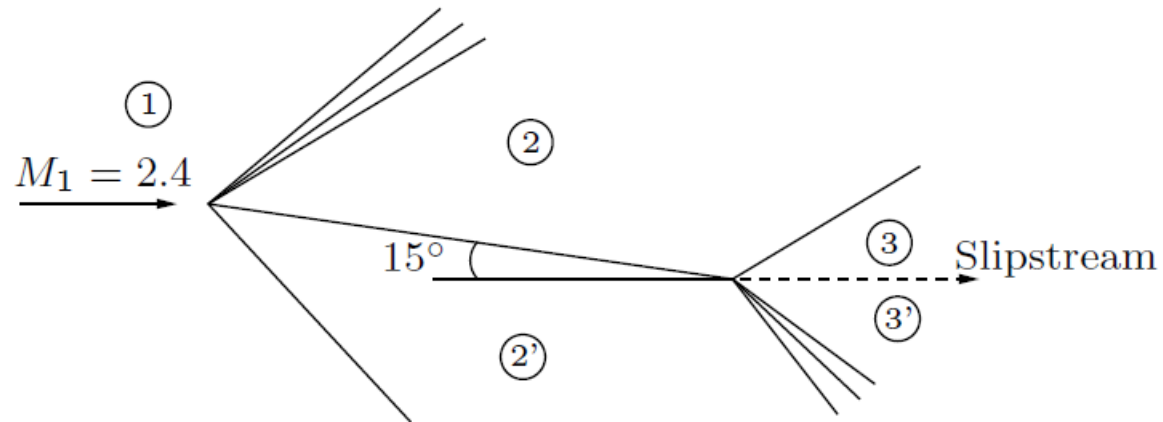
Region	M	ν (°)	μ (°)	p/p_{01}	T/T_{01}
1	2.40	36.8	24.6	0.0684	0.465
2	2.62	41.8	22.5	0.0486	0.421
3	3.71	61.8	15.6	0.0098	0.267
4	2.00	26.5	30.0	0.0707	0.555
2'	1.31	6.5	49.7	0.2736	0.745
3'	2.00	26.5	30.0	0.0986	0.555
4'	2.19	31.5	27.1	0.0689	0.501

With slip stream

Region	M	ν	μ	p/p_{01}	T/T_{01}
1	2.40	36.8°	24.6°	0.0684	0.465
2	2.62	41.8°	22.5°	0.0486	0.421
3	3.71	61.8°	15.6°	0.0098	0.267
4	2.00	26.5°	30.0°	0.0707	0.555
2'	1.31	6.5°	49.7°	0.2736	0.745
3'	2.00	26.5°	30.0°	0.0986	0.555
4'	2.23	32.5°	26.6°	0.0689	0.501

26 deg(top),6 deg(bot) to
get back to free stream
direction

A flat plate is kept at 15° angle of attack to a Mach 2.4 air stream. Solve the flow field around the plate and determine the flowfield properties using shock-expansion theory.



Region	M	ν	μ	p/p_{01}	T/T_{01}
1	2.4	36.8°	24.6°	0.0684	0.465
2	3.11	51.8°	18.8°	0.0231	0.341
3	2.33	35.0°	25.4°	0.0675	0.480
2'	1.80	20.7°	33.8°	0.1629	0.607
3'	2.36	35.7°	25.1°	0.0679	0.473

Consider an infinitely thin flat plate at a 5° angle of attack in a Mach 2.6 free stream. Calculate the lift and drag coefficients.

$$\nu_2 = \nu_1 + \alpha = 41.41^\circ + 5^\circ = 46.41^\circ$$

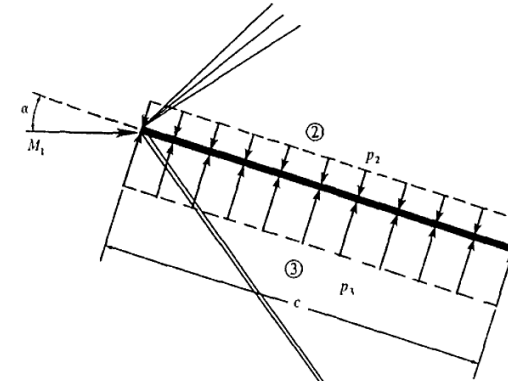
$$M_2 = 2.85,$$

$$\frac{p_{02}}{p_2} = 29.29$$

$$\beta = 26.5^\circ$$

$$\frac{p_2}{p_1} = \frac{p_2}{p_{02}} \cdot \frac{p_{02}}{p_0} \cdot \frac{p_{01}}{p_1} = \frac{1}{29.29} \cdot 1 \cdot (19.95) = 0.681$$

$$\frac{p_3}{p_1} = 1.403$$



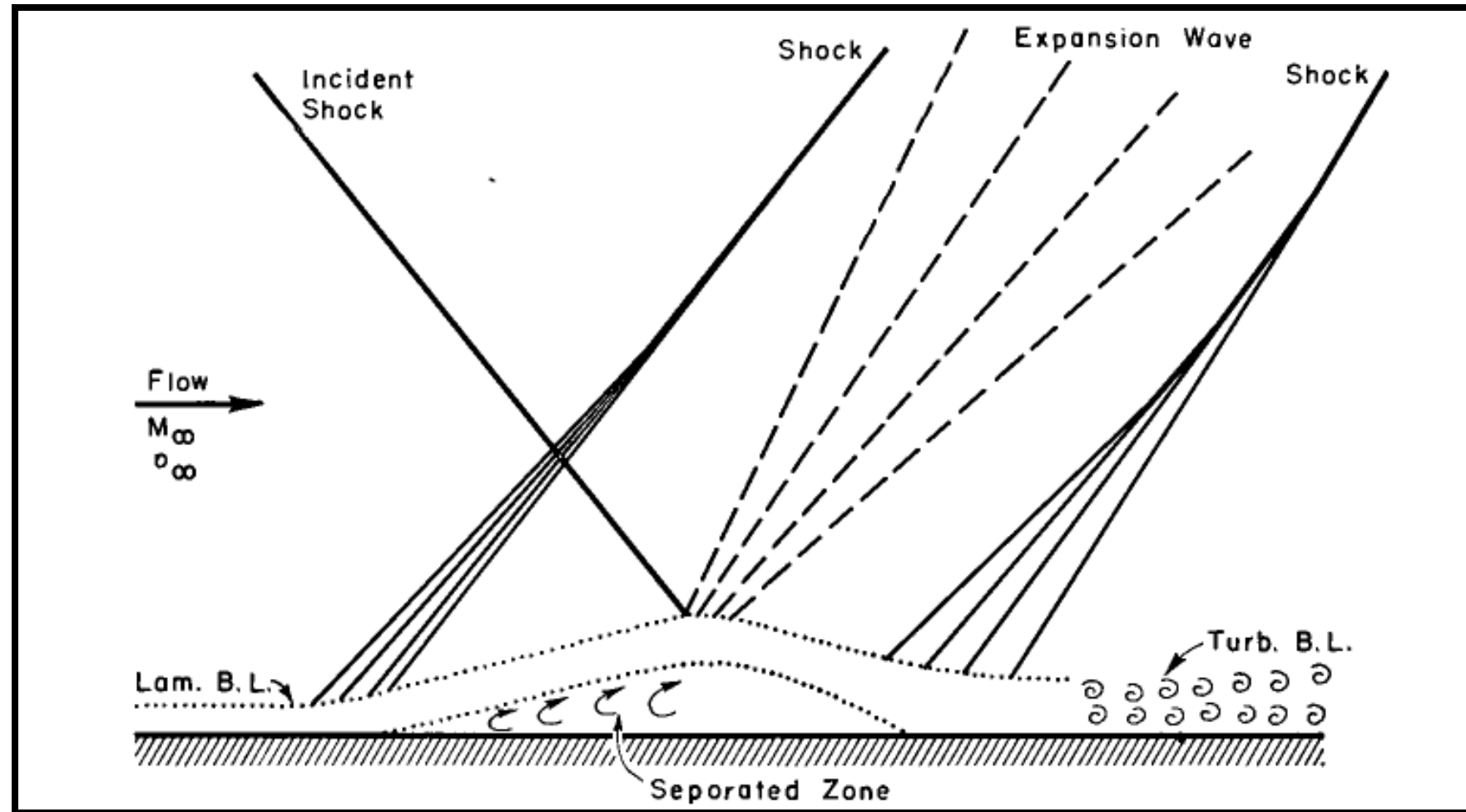
$$c_l = \frac{L'}{q_t c} = \frac{2}{\gamma M_1^2} \left(\frac{p_3}{p_1} - \frac{p_2}{p_1} \right) \cos \alpha$$

$$c_l = \frac{2}{(1.4)(2.6)^2} (1.403 - 0.681) \cos(5^\circ) = \boxed{0.152}$$

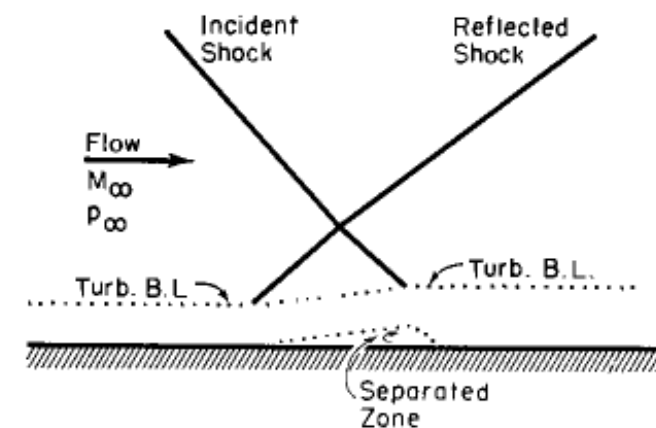
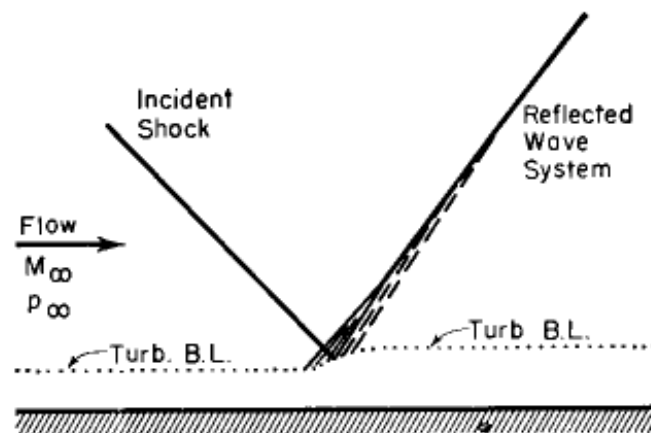
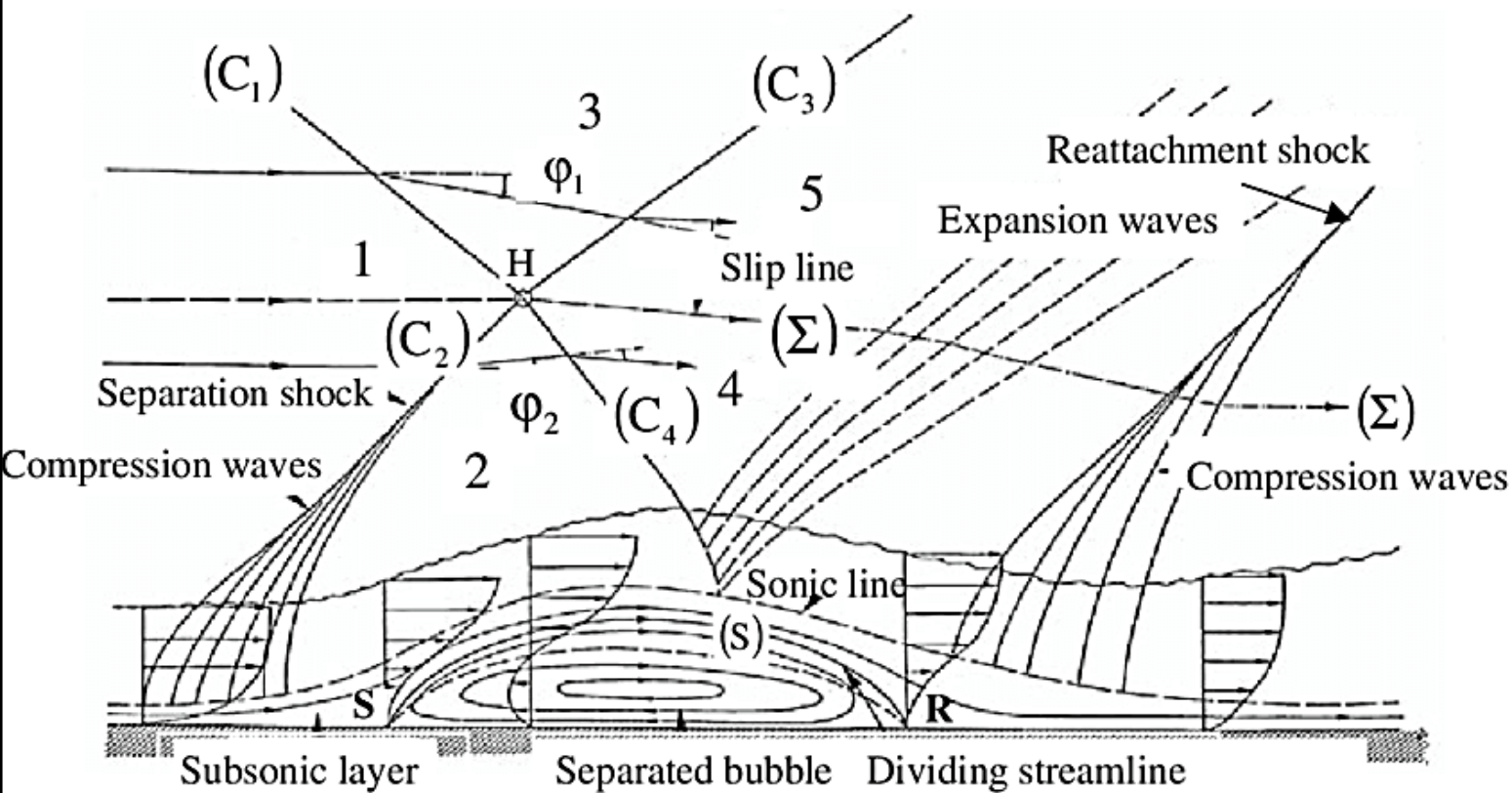
$$c_d = \frac{D'}{q_t c} = \frac{2}{\gamma M_1^2} \left(\frac{p_3}{p_1} - \frac{p_2}{p_1} \right) \sin \alpha$$

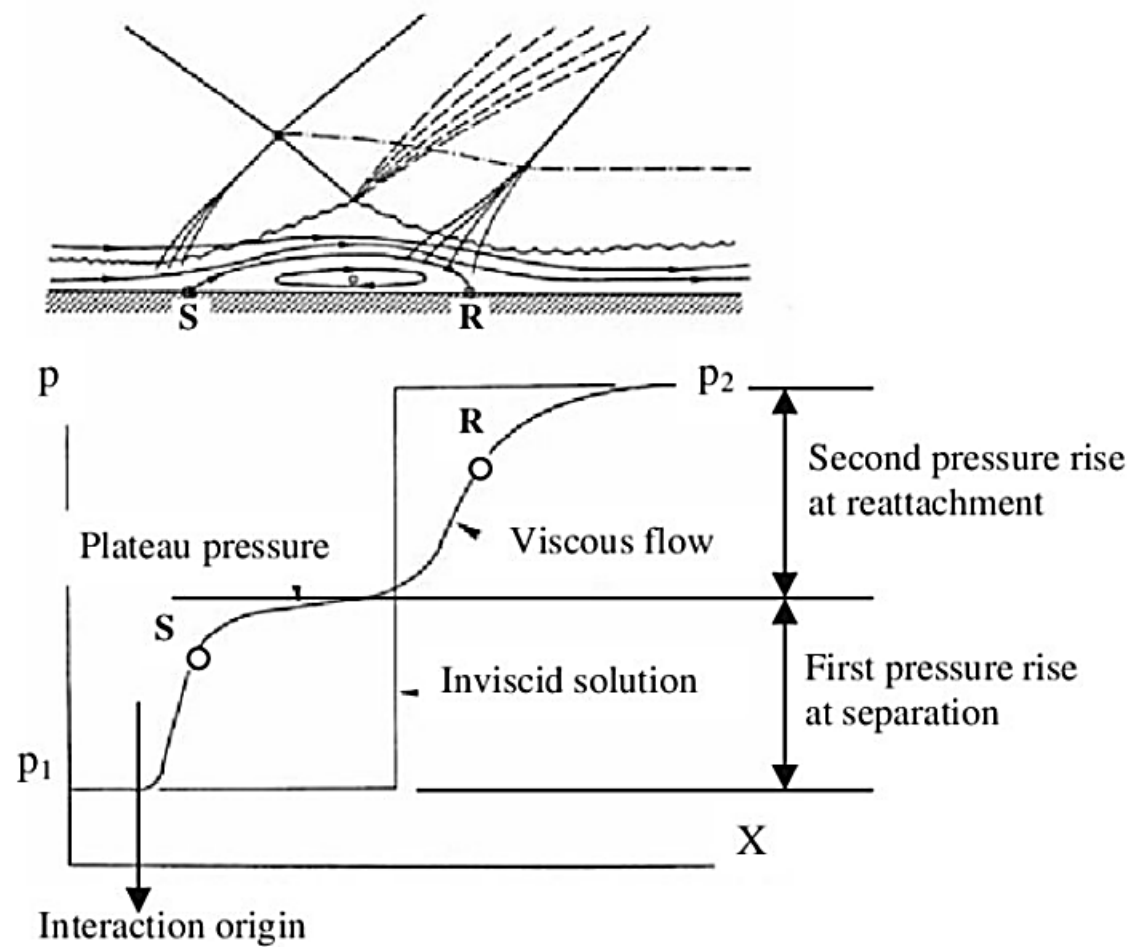
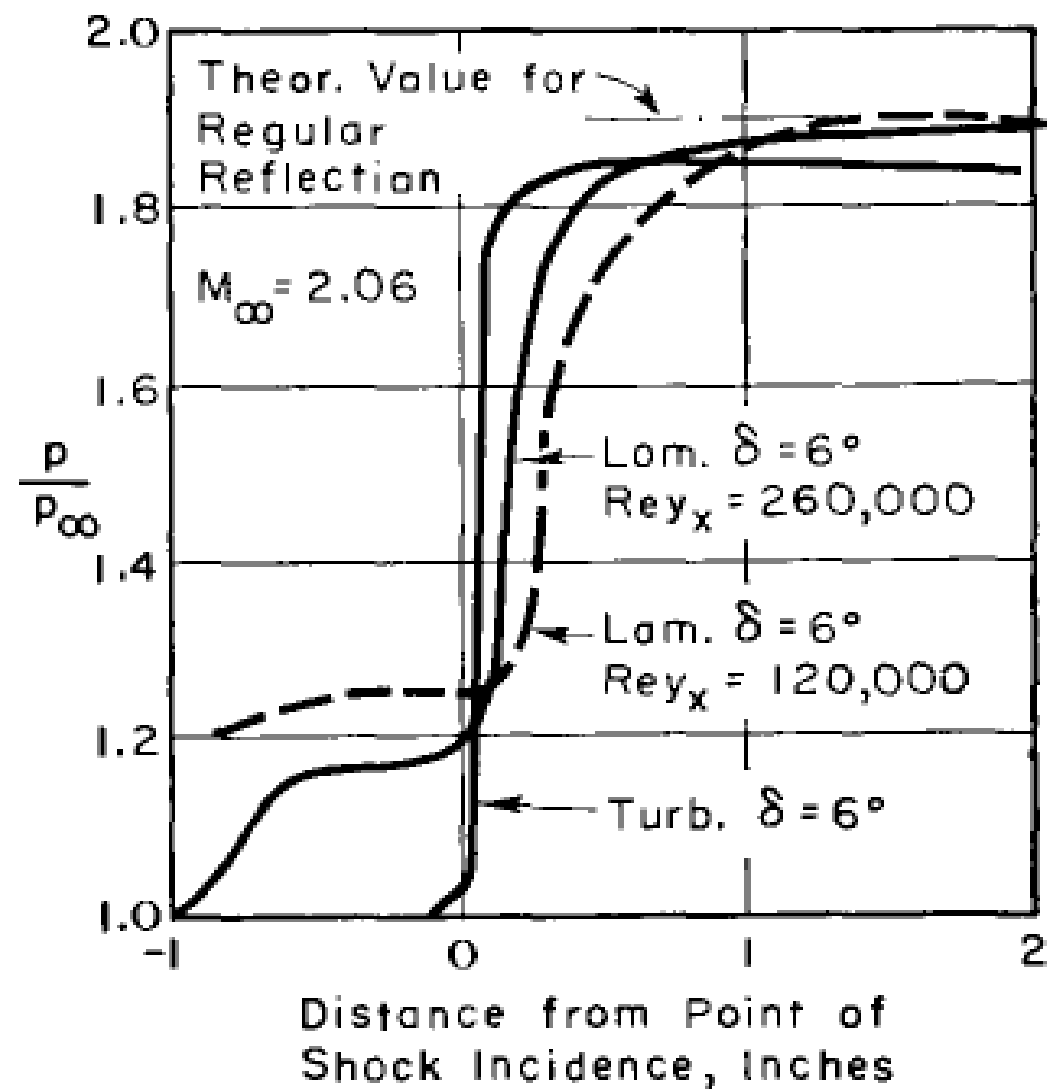
$$c_d = \frac{2}{(1.4)(2.6)^2} (1.403 - 0.681) \sin(5^\circ) = \boxed{0.0133}$$

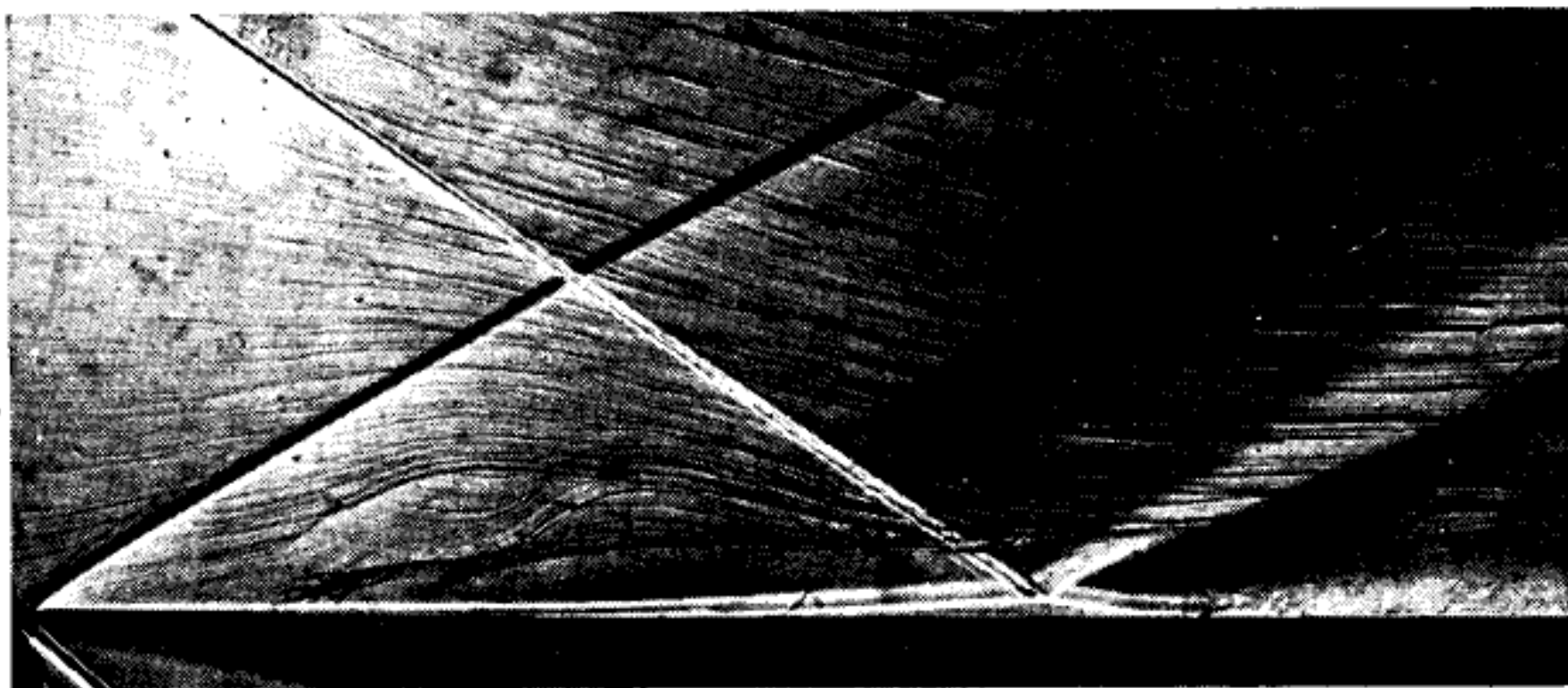
Shockwave boundary layer interaction -SBLI



SBLI For a Turbulent B.Layer

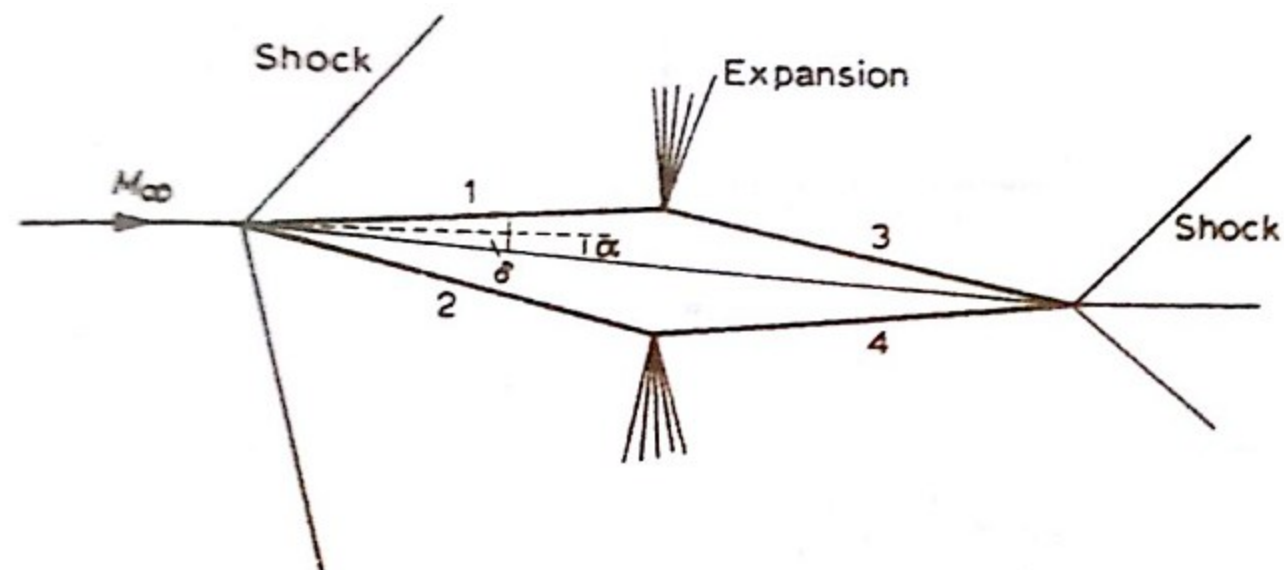






[Simulation](#)

Alpha < delta

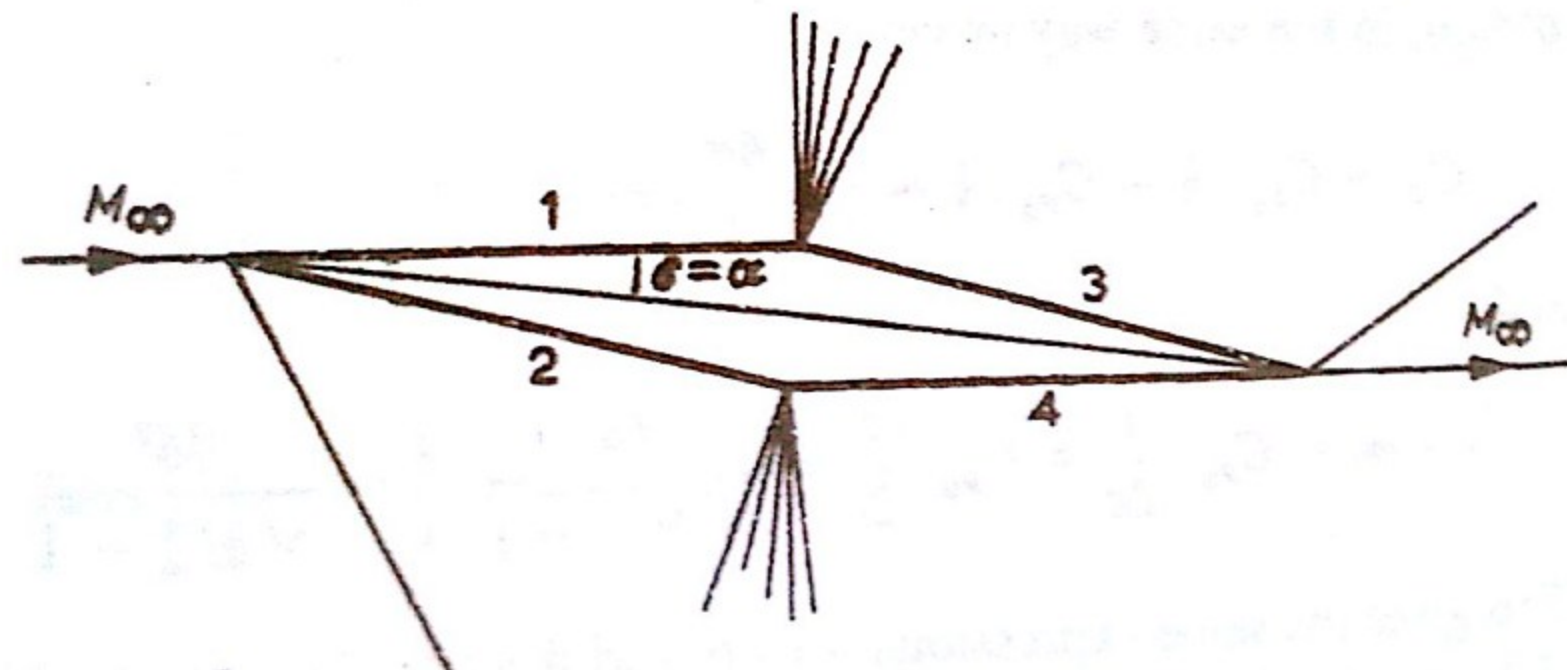


$$C_{p_1} = \frac{2(\delta - \alpha)}{\sqrt{M_\infty^2 - 1}},$$

$$C_{p_2} = \frac{2(\delta + \alpha)}{\sqrt{M_\infty^2 - 1}},$$

$$C_{p_3} = -\frac{2(\delta + \alpha)}{\sqrt{M_\infty^2 - 1}},$$

$$C_{p_4} = -\frac{2(\delta - \alpha)}{\sqrt{M_\infty^2 - 1}}.$$



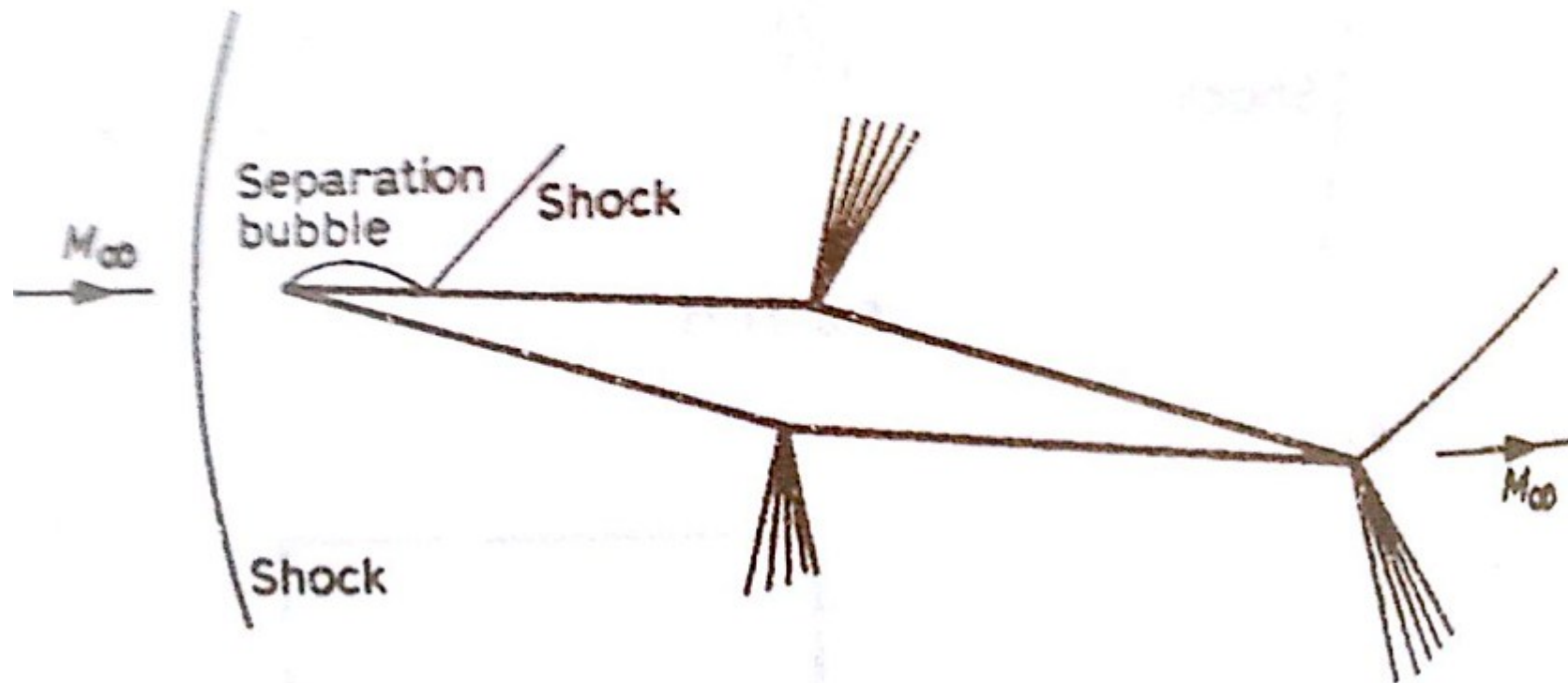
$$C_{p_1} = 0, \quad C_{p_2} = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}},$$

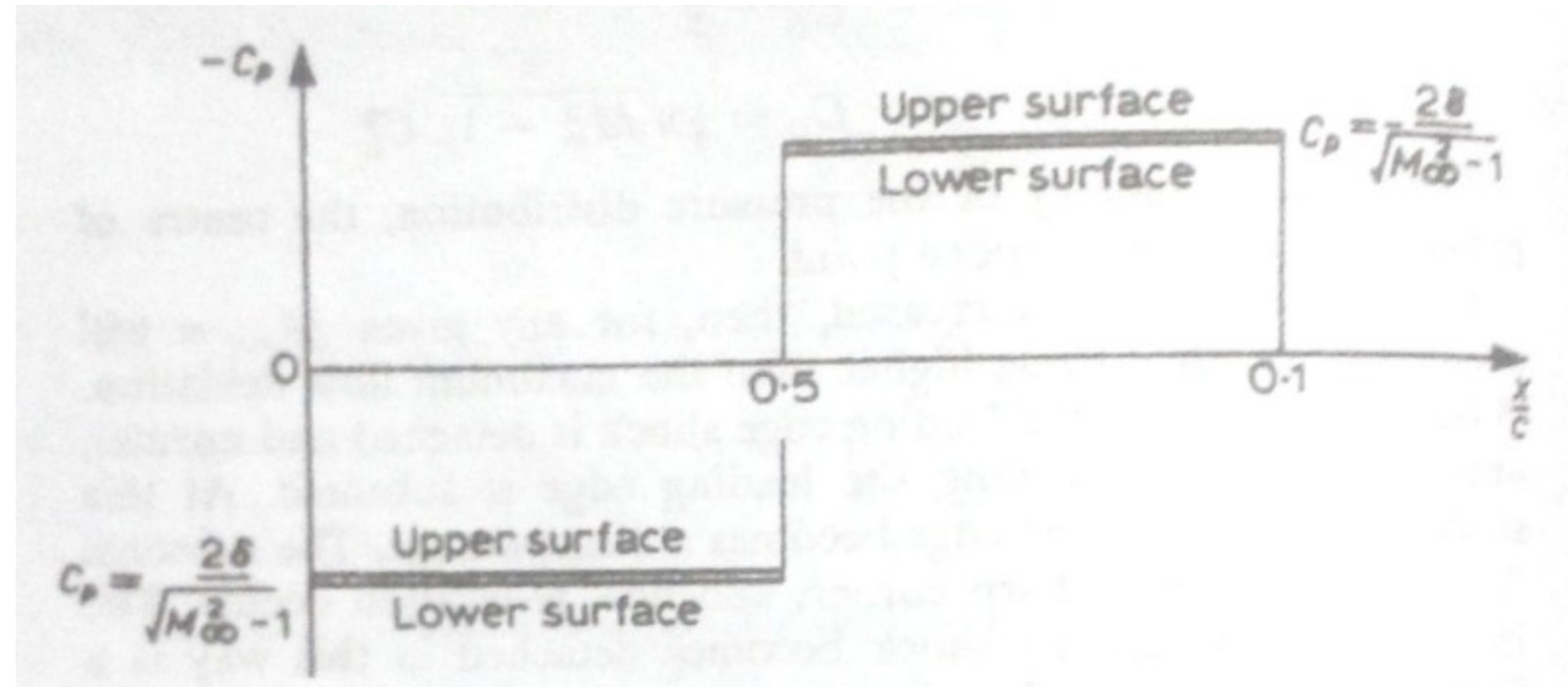
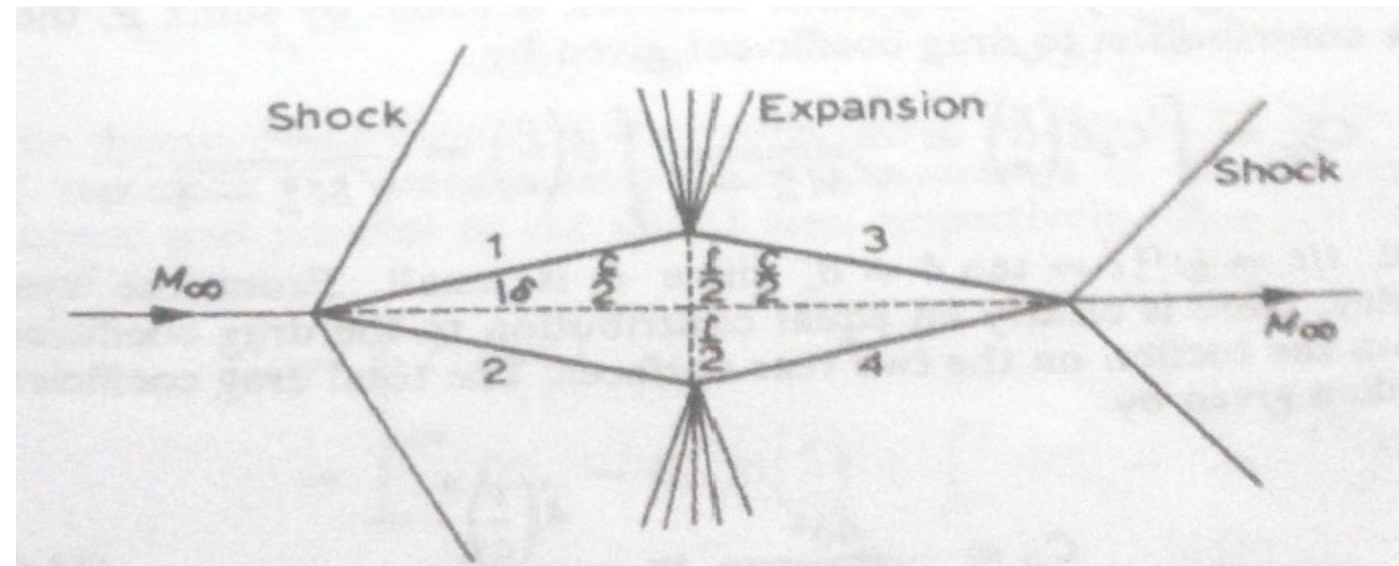
$$C_{p_3} = -\frac{4\alpha}{\sqrt{M_\infty^2 - 1}}, \quad C_{p_4} = 0$$

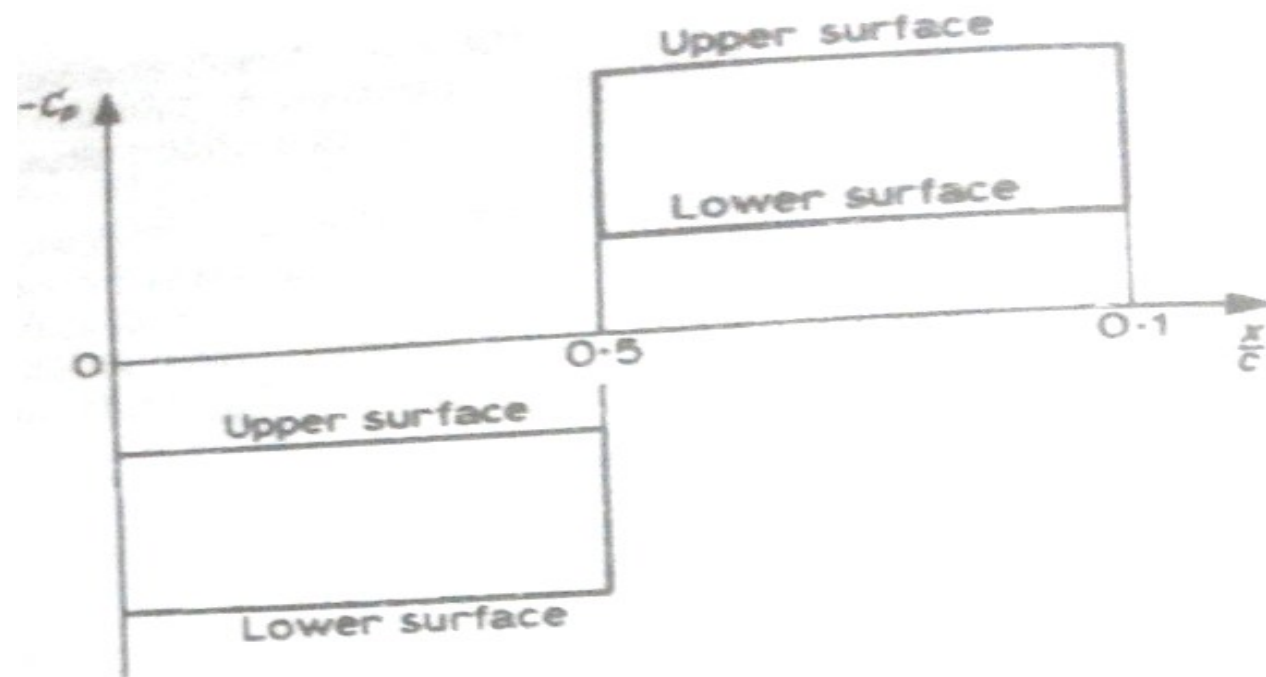
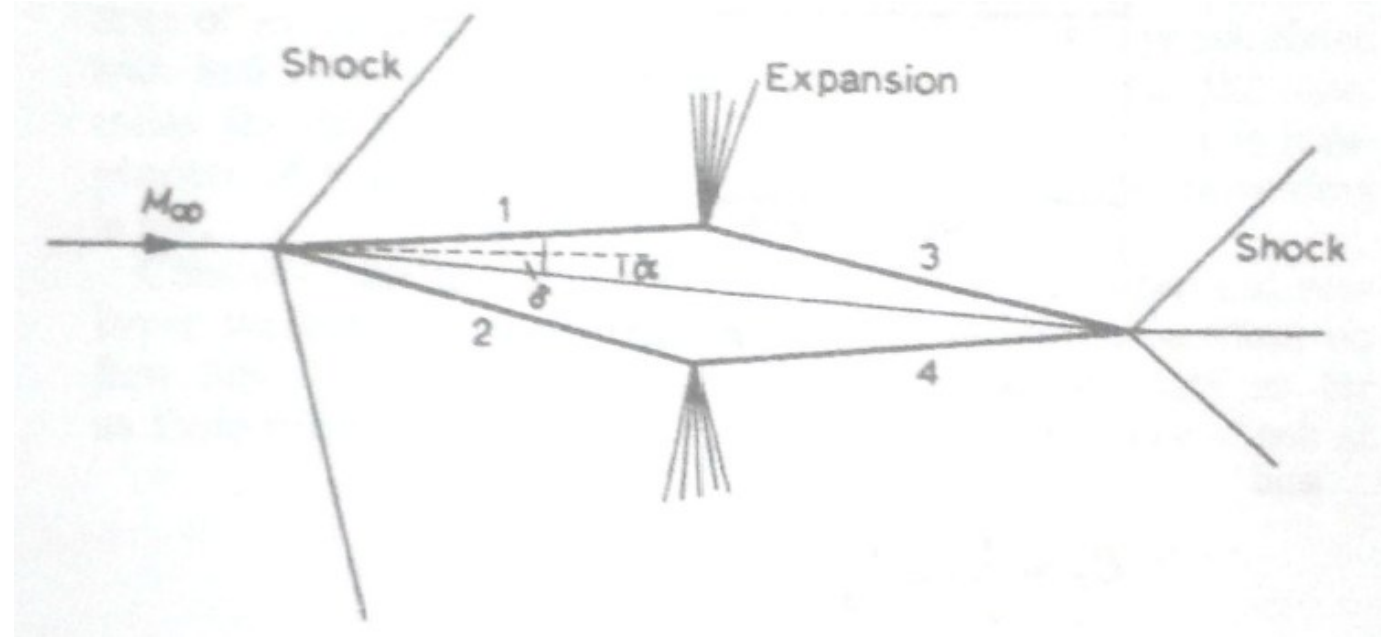
$$\alpha > \delta$$

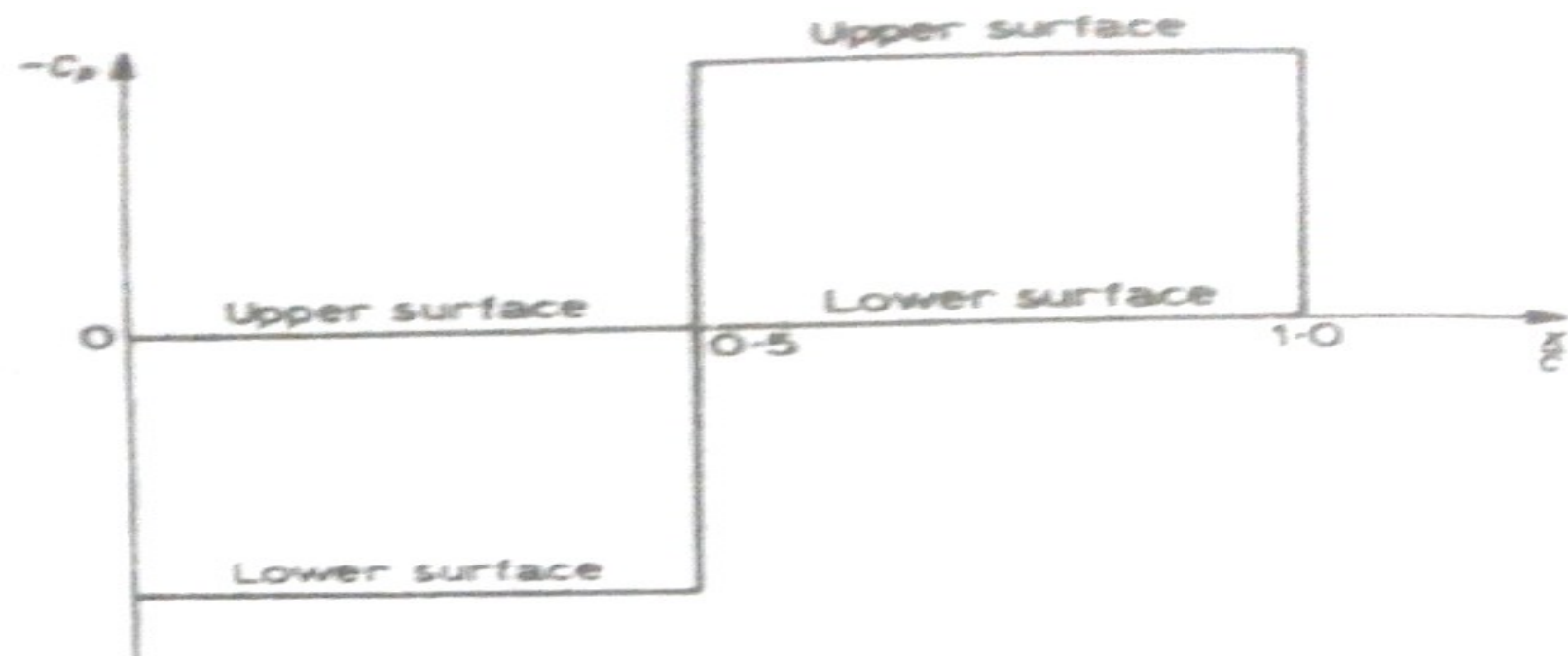
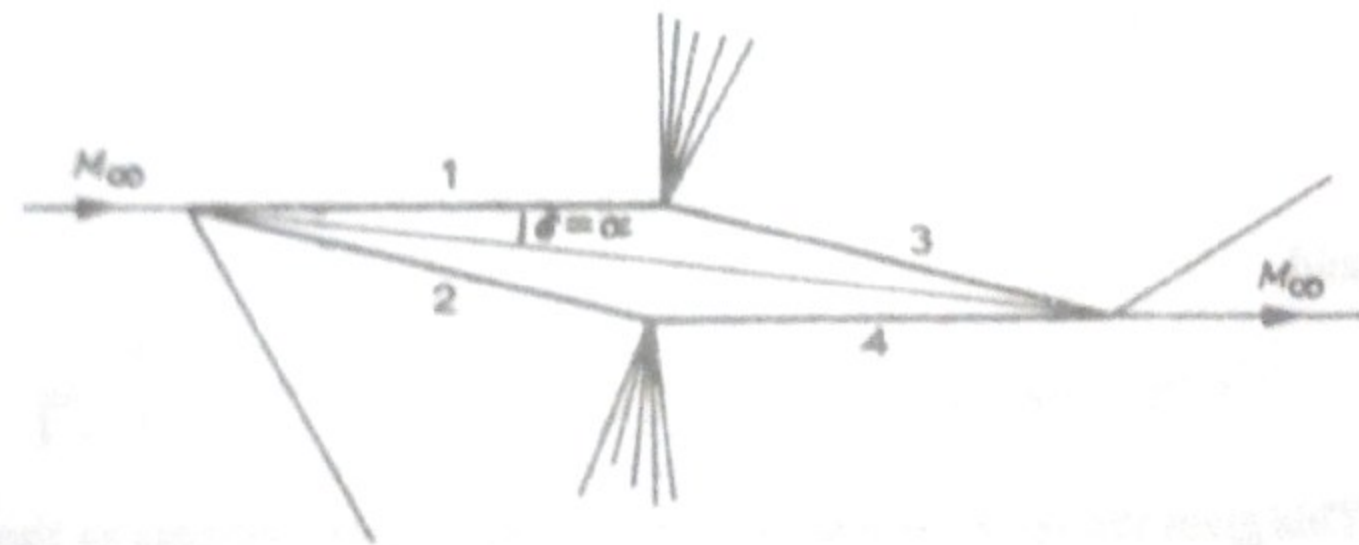
$$C_{p_1} = -\frac{2(\alpha - \delta)}{\sqrt{M_\infty^2 - 1}} = \frac{2(\delta - \alpha)}{\sqrt{M_\infty^2 - 1}}, \quad C_{p_2} = \frac{2(\alpha + \delta)}{\sqrt{M_\infty^2 - 1}}$$

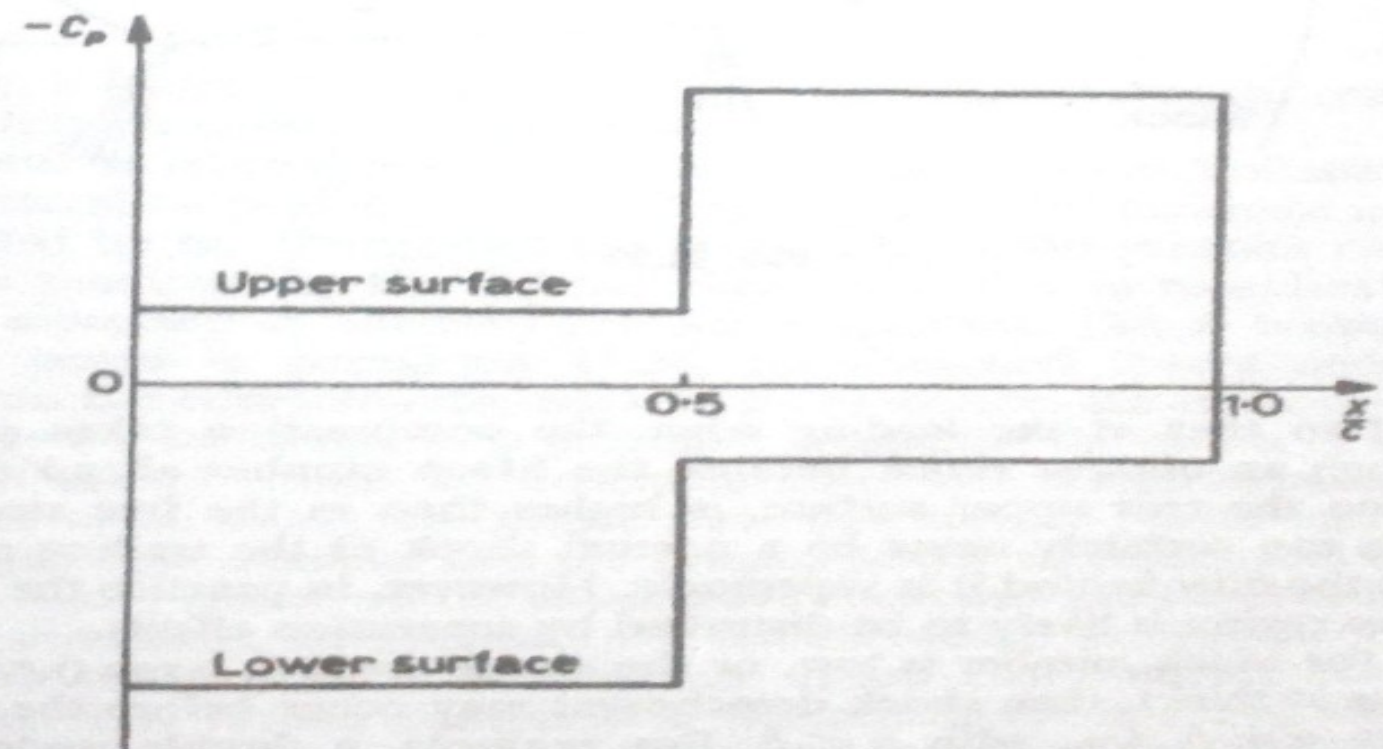
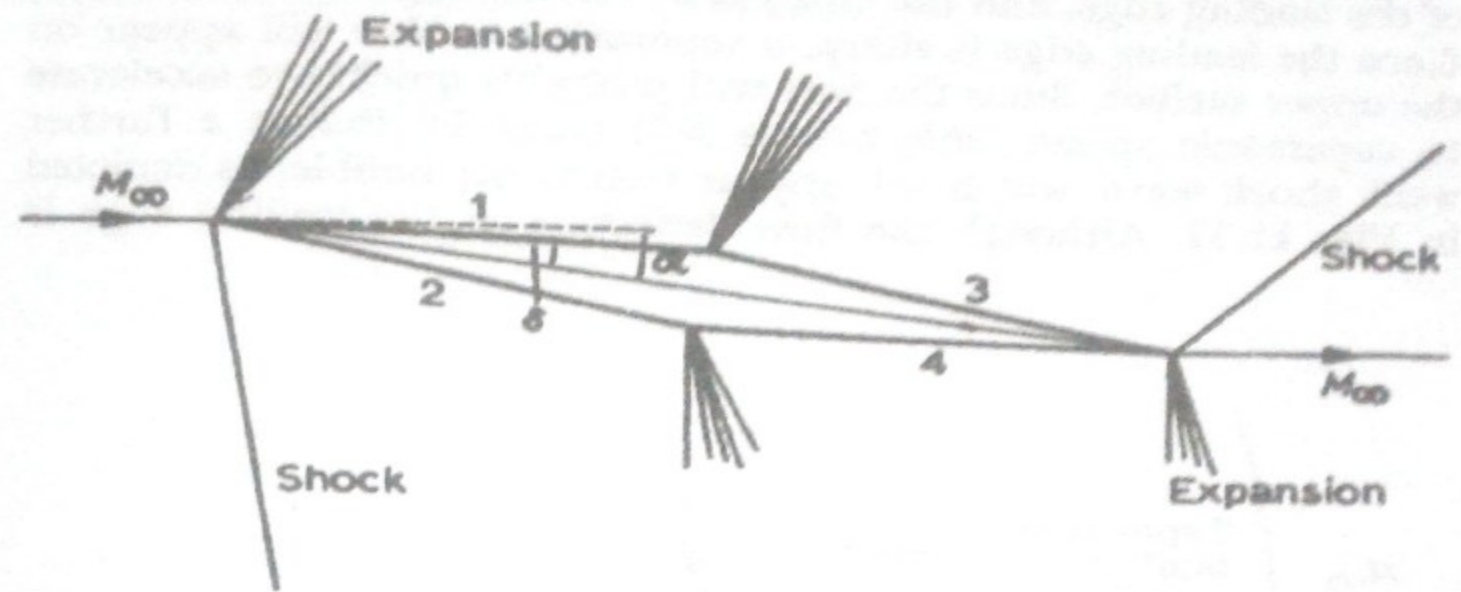
$$C_{p_3} = -\frac{2(\alpha + \delta)}{\sqrt{M_\infty^2 - 1}}, \quad C_{p_4} = \frac{2(\alpha - \delta)}{\sqrt{M_\infty^2 - 1}} = -\frac{2(\delta - \alpha)}{\sqrt{M_\infty^2 - 1}}$$

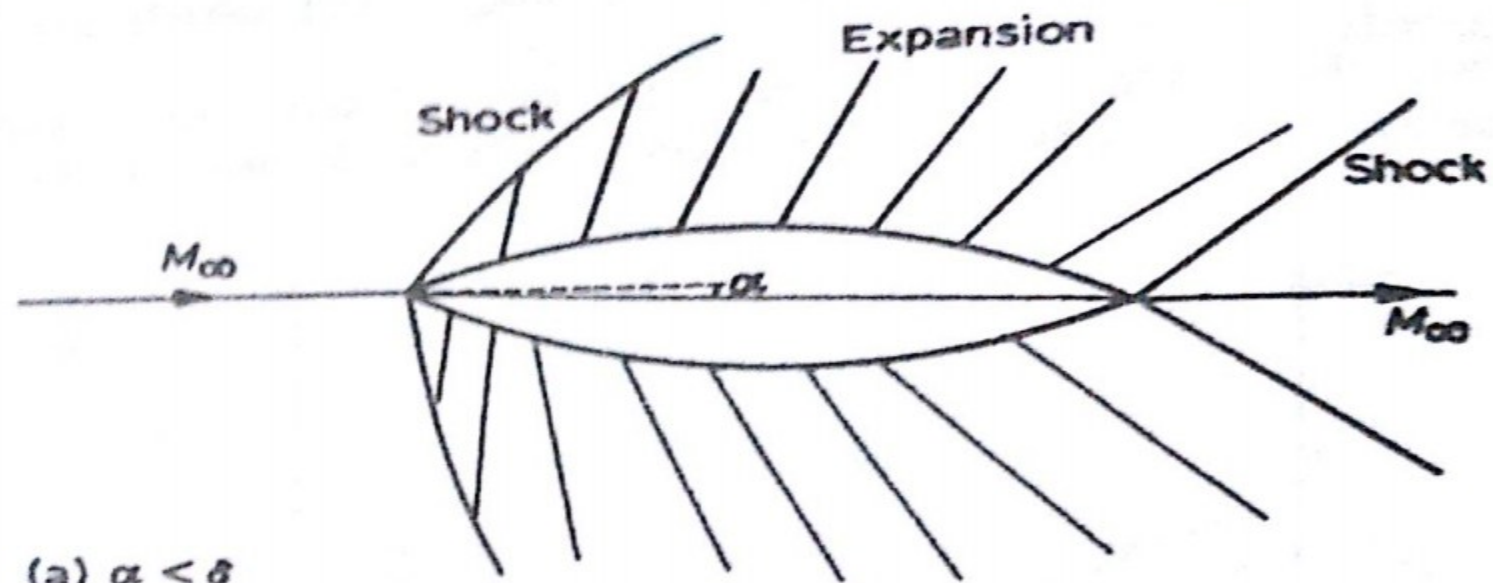




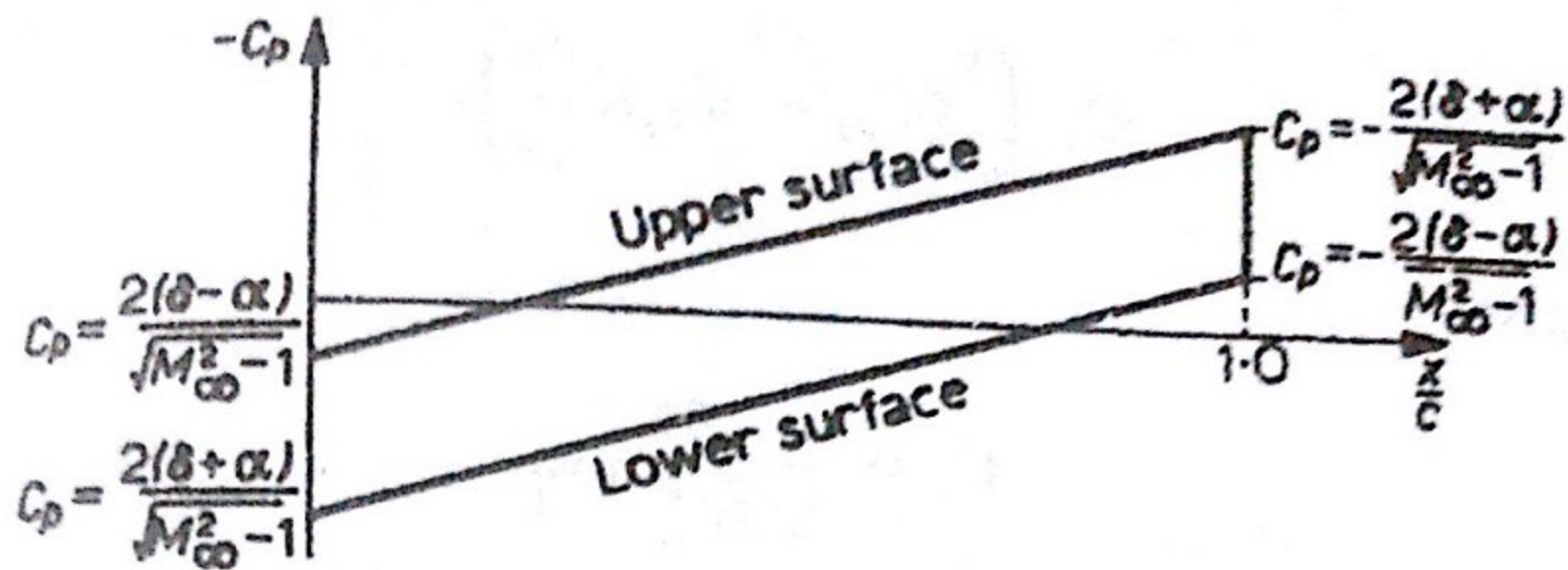








(a) $\alpha < \delta$



(a) $\alpha < \delta$

